

QUATERNION LATTICE POINTS NEAR A GREAT CIRCLE AT A SINGLE 2-POWER NORM: A SPARSITY QUESTION

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ABSTRACT. We isolate a lattice-point question that decides an exchange rate between classical memory and magic states in quantum compilation. The question is whether quaternions of a fixed 2-power norm in the maximal order of $(-1, -1)_{\mathbb{Q}(\sqrt{2})}$ can occupy almost every ε -cell along some great circle of S^3 at the scale where the determinant method is exactly saturated. We state what we can prove, why the methods we know stop there, and the precise questions we would like to ask. Background and all proofs are in arXiv:2609.37368.

1. SETTING

Let $K = \mathbb{Q}(\sqrt{2})$, $D = (-1, -1)_K$, and let

$$\mathcal{O} = \mathcal{O}_K \oplus \mathcal{O}_K \frac{1+i}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+j}{\sqrt{2}} \oplus \mathcal{O}_K \frac{1+i+j+k}{2}$$

be its maximal order. The single-qubit Clifford+ T group modulo phase is $\Gamma = \mathcal{O}[1/\sqrt{2}]^\times / \mathcal{O}_K[1/\sqrt{2}]^\times$, a lattice in $\mathrm{PU}(2) \times \mathrm{PU}(2) \times \mathrm{PGL}_2(K_P)$ with $P = (\sqrt{2})$. The quaternion algebra is definite at both real places, so Γ acts on the 3-regular tree X_P with vertex stabiliser the Clifford group $\cong S_4$. The T -count of a word equals its tree distance from the root. Let Λ_t be the set of elements of T -count $\leq t$. Then $|\Lambda_t| = 72 \cdot 2^t - 48$.

Numerators. An element of minimal T -count t is $\pm w / |\sigma_1(w)|$ for some $w \in \mathcal{O}$ with $\mathrm{nrd}(w) = n_t$, where $n_t \in \{2^{t/2}, (2 + \sqrt{2})2^{(t-1)/2}\}$. Thus $\sigma_1(w)$ and $\sigma_2(w)$ lie on 3-spheres of radii $\asymp R' := 2^{t/4}$, and $\#\{w\} \asymp R'^4$.

Tubes. A *frame* is a pair $G_1, G_2 \in \mathrm{SU}(2)$, and $\mathcal{C} = G_1\{R_z(\theta)\}G_2$ is a great circle of $S^3 = \mathrm{SU}(2)$. With $d(A, B) = \inf_\phi \|A - e^{i\phi}B\|$, put

$$N_t(\mathcal{C}, \varepsilon) := \#\{W \in \Lambda_t : \exists \theta, d(W, G_1 R_z(\theta) G_2) \leq \varepsilon\}.$$

In arithmetic terms this counts the $w \in \mathcal{O}$ with $\mathrm{nrd}(w) = n_t$ whose σ_1 -image lies within Euclidean distance $\asymp \varepsilon R'$ of the great circle $R'\mathcal{C}$. There is *no condition* on $\sigma_2(w)$. In Hopf form, N_t counts the W with $\angle(R(W)x_2, x_1) \leq 4 \arcsin(\varepsilon/2)$, where $x_1 = G_1 \hat{z}$ and $x_2 = G_2^{-1} \hat{z}$, and $R(W) \in \mathrm{SO}(3)$. The Haar expectation is $\asymp 2^t \varepsilon^2$.

Cells. Fix a grid $\{G_1 R_z(2\pi u/Q) G_2\}_{u \in \mathbb{Z}_Q}$ on $\mathcal{C}$ with $Q \asymp 1/\varepsilon$ and spacing $> 2\varepsilon$, and put

$$M_t(\mathcal{C}, \varepsilon) := \#\{u \in \mathbb{Z}_Q : \exists W \in \Lambda_t, d(W, G_1 R_z(2\pi u/Q) G_2) \leq \varepsilon\},$$

the number of *occupied* ε -cells. Trivially $M_t \leq \min(Q, N_t)$.

2. WHAT IS KNOWN

Write $\varepsilon = R'^{-a_0}$, so $t \asymp (4/a_0) \log_2(1/\varepsilon)$.

- (R) *Spectral.* The Ramanujan bound for Γ (Lubotzky–Phillips–Sarnak, Parzanchevski–Sarnak), via a Poisson-kernel majorant of the cap in Hopf form, gives $N_t \leq C_1 2^t \varepsilon^2 + C_2(t+1)2^{t/2}$ for all frames. The error is $2^{t/2} = R'^2$, which exceeds the cell count $Q \asymp R'^{a_0}$ for every $a_0 < 2$.
- (D) *Determinant method* (arXiv:2609.37368, Thm. 8 and Apps. E–H).

- Five numerators in a box of core length $\ell R'$ have a determinant in $\frac{1}{16}\mathbb{Z}[\sqrt{2}]$ of size $\lesssim \ell^3 \varepsilon^2 R'^4$ in σ_1 and $\lesssim R'^4$ in σ_2 .
- So the points of a box are cohyperplanar once $\ell^3 \varepsilon^2 R'^8 < c$.
- Combined with heights of the resulting sphere sections, rational projections, and an exact SMT check of the continuum case analysis, this gives, for every frame and every $0 < \mu \leq 1/10$,

$$N_t(\mathcal{C}, \varepsilon) \leq R'^{8/5-4\mu/11+o(1)} \quad \text{whenever } \varepsilon \leq R'^{-(8/5+\mu)}.$$

- At the critical scale $\varepsilon \asymp R'^{-8/5} = 2^{-2t/5}$ the determinant boxes are exactly ε -cells, and the bound is $N_t \leq \varepsilon^{-1} R'^{o(1)}$: *one point per cell*. The volume is only $R'^{4/5} = \varepsilon^{-1/2}$.
- (C) *Clifford frames*. If G_1, G_2 are Cliffords, the off-diagonal entry is an arithmetic coordinate. The divisor bound in $\mathbb{Q}(\zeta_8)$ then gives $N_t \leq 2^{O(t/\log t)}(t+1+\varepsilon^{2t})$, the volume law. For $G_1, G_2 \in \Gamma$ of T -count $\leq h$ one loses a factor 2^h .
- (N) *Numerics*. Exhaustive enumeration to $t = 22$ is consistent with “volume + a sparse cluster”:
- The cluster consists of powers of one short hyperbolic word along its axis. It forces a term linear in t .
 - Heuristically, with exact instances, balls of radius $\varepsilon = 2^{-t/\alpha}$ around low-height Hecke points $g/|g|$ carry $2^{t(2/3-5/(3\alpha))}$ points on a single sphere section. So covering a tube by balls should not beat the critical scale, and tubes have to be treated globally.

3. THE QUESTION

Question (Q1: any power saving at the critical scale). *Is there $\kappa > 0$ such that $M_t(\mathcal{C}, 2^{-2t/5}) \leq 2^{(2/5-\kappa)t}$ for all great circles $\mathcal{C}$ and all large t ? In words: along every great circle, is a power-fraction of the ε -cells empty at $\varepsilon = 2^{-2t/5}$?*

Question (Q2: cell form of the volume law). *Is $M_t(\mathcal{C}, \varepsilon) \leq 2^{o(t)}(1 + 2^t \varepsilon^2)$ uniformly in $\mathcal{C}$ and ε ? It suffices to have $M_t \leq 2^{t/3+o(t)}$ whenever $\varepsilon \geq 2^{-t/3}$.*

An affirmative answer to Q1 (in a small neighbourhood of the critical scale) raises our unconditional exchange rate above $5/2$. An affirmative answer to Q2 gives the conjectured value 3. The questions are *exactly* what the application needs. The following short argument is not in the paper.

Proposition 1 (two rounds). *For two-round streams the optimal exchange rate (committed T gates per bit of memory forgone) is, up to $o(1)$, the largest α such that*

$$\max_{\mathcal{C}} M_t(\mathcal{C}, \varepsilon) \leq 2^{t/\alpha+o(t)} \quad \text{for all } t < \alpha \log_2(1/\varepsilon).$$

Only the frames with $G_1 = I$ enter. Equivalently, the relevant tubes are the ε -neighbourhoods of Hopf fibres, and M_t counts the phase cells of Clifford+ T states $W^{-1}|0\rangle$ whose Bloch vectors lie in an ε -cap.

Sketch. Lower bound.

- Given everything except the share a of round 1, the committed word is an injective function of a into the cells of one tube.
- That tube is determined by the round-2 output, which is free, so the frame is arbitrary.
- A Gibbs-measure entropy bound with the cell profile above gives the rate.

Upper bound.

- Let $U \subset \mathbb{Z}_Q$ be the occupied cells at cost t for some frame.
- Choose public shifts K with $\bigcup_{k \in K} (U - k) = \mathbb{Z}_Q$ and $|K| \leq (Q/|U|) \ln Q + 1$.
- For the share a , pick $k \in K$ with $a + k \in U$. Commit the word of cell $a + k$, which costs $\leq t$, and store k .
- In round 2, emit an approximation of $G_2^{-1} R_z(\Delta(a_2 - k))$. This costs no committed magic.

- The memory saved is $\log_2 |U| - \log_2 \ln Q - O(1)$ bits at cost t . No condition on the gaps of U is needed.

□

So no averaging or modelling device avoids the worst-case tube. For more than two rounds the later rounds have arithmetic left frames $G_1 \in \Gamma$ of large height. The relevant points are then the orbit $\Lambda_t \cdot \phi$ of the arithmetic state $\phi = G_1^{-1}|0\rangle$.

4. WHY WE BELIEVE THE METHODS WE KNOW STOP AT THE CRITICAL SCALE

(1) *One prime.*

- All norms are powers of P .
- In Marshall's geodesic-restriction work (Duke 2016) the single-norm count of Hecke returns near a torus (his Lemma 3.2) is elementary. The volume law is obtained only on average over $m \sim M$ (Prop. 5.2), via $\sum_m g(m/M) \lambda_\pi(m) \ll M^{-A}$ from the continuation of $L(s, \pi)$.
- At a single prime the analogous generating function $\sum_k \lambda_\pi(P^k) X^k = (1 - a_\pi X + 2X^2)^{-1}$ has, under Ramanujan, its poles on $|X| = 2^{-1/2}$. So no such gain is available, and the spectral error $2^{t/2}$ appears to be intrinsic.
- Equivalently, the entropy sum $\sum_t N_t 2^{-t/\alpha}$ needed for rate $\alpha > 2$ is evaluated beyond the radius of convergence of every non-trivial local factor.

(2) *The determinant method is saturated.* At $\varepsilon = R'^{-8/5}$ the only configuration the case analysis cannot exclude is one lattice point per ε -cell. In that configuration:

- Consecutive points differ by $d \in \mathcal{O}$ with $|\sigma_1 d| \lesssim \varepsilon R'$ and $|\sigma_2 d| \lesssim R'$.
- This product of balls contains $\asymp \sum_{N(m) \lesssim R'^{4/5}} r_{\mathcal{O}}(m) \asymp R'^{8/5} \asymp \varepsilon^{-1}$ lattice vectors, exactly the number of cells.
- Each difference fixes its cell up to $O(1)$.

So full occupancy is not contradicted at the level of differences. Five points in J consecutive cells have a determinant of norm $O(J^3)$, so there is no gap principle. A leading-order count suggests that higher-degree auxiliary forms on the quadric $\text{nrd} = n_t$ do worse, because the curvature enters only through monomials linear in the normal coordinate.

(3) *A congruence form (heuristic).*

- For Hopf fibres, choose a Dirichlet approximation $p/q \in \mathbb{Q}(\zeta_8)$ to the fibre's stereographic coordinate, with $H = |p|^2 + |q|^2$ and $N(H) \asymp \varepsilon^{-1}$.
- This turns the count exactly into a Clifford-frame count, where the divisor bound applies. The count is of $(\lambda, g) \in \mathbb{Z}[\zeta_8]^2$ with $|\lambda|^2 + |g|^2 = H n_t$, g in a small box, and one congruence $\lambda \equiv \kappa g \pmod{H}$.
- After rebalancing by units, the box covers about $q^{3/4}$ of the $q = N(H)^2$ residue classes of g .
- An error of Pólya–Vinogradov size $q^{1/2} = \varepsilon^{-1}$ is exactly the critical exponent above. Square-root cancellation in the incomplete sum would give Q2.

5. WHAT WE WOULD LIKE TO ASK

- (a) Is any single-norm bound for $\#\{\gamma \in R(n) : \gamma \text{ within } \varepsilon \text{ of a generic torus coset}\}$ beyond geometry of numbers known? We mean a definite quaternion order, prime-power n , and microscopic ε , over any number field.
- (b) Could theta-lift or fourth-moment methods (as in Khayutin–Nelson–Steiner, or the S^3 sup-norm work) be adapted to a *fixed* prime-power norm? It would also suffice to have them with an average over a family of tori that is uniform enough to control the worst torus at the critical scale.

- (c) A toy model over $\mathbb{Q}$. Take $x \in \mathbb{Z}^4$ (or Hurwitz quaternions) with $|x|^2 = n = p^k$, within distance $n^{1/10}$ of a 2-plane through the origin.
- The five-point determinant gives $\leq n^{2/5+o(1)}$: one point per cell of length $n^{1/10}$.
 - The volume is $n^{1/5}$.
- Is a bound $n^{2/5-\kappa}$ known, or expected to be within reach?
- (d) Conversely, is there a known mechanism producing a great circle with $n^{2/5-o(1)}$ such points? That would make our 5/2 sharp by Proposition 1.

REFERENCES

- [1] J. Yang, Y. Li, X.-H. Deng, *A memory-magic exchange law in streaming Clifford+T compilation*, arXiv:2609.37368.
- [2] A. Lubotzky, R. Phillips, P. Sarnak, *Hecke operators and distributing points on the sphere I*, Comm. Pure Appl. Math. 39 (1986).
- [3] O. Parzanchevski, P. Sarnak, *Super-Golden-Gates for $PU(2)$* , Adv. Math. 327 (2018).
- [4] S. Marshall, *Geodesic restrictions of arithmetic eigenfunctions*, Duke Math. J. 165 (2016).
- [5] I. Khayutin, P. D. Nelson, R. S. Steiner, *Theta functions, fourth moments of eigenforms and the sup-norm problem II*, Forum Math. Pi 12 (2024).
- [6] R. S. Steiner, *Sup-norm of Hecke–Laplace eigenforms on S^3* , Math. Ann. 377 (2020).
- [7] J. Bourgain, Z. Rudnick, *Restriction of toral eigenfunctions to hypersurfaces and nodal sets*, GAFA 22 (2012).

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